

Incorrect Indication of Power Supply Fault and Engine Shutdown Unit

Symptoms

- Alarm is sounding on DCU410 unit with false alarm power supply +24-VDC fault.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The DCU410 unit alarm will sound if the power supply +24-VDC fault is detected.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check SDU410 unit.	
	STEP 1A. Check SDU410 unit for illuminated LEDs.	Is power LED illuminated?
STEP 2.	Check SDU410 unit power supply.	
	STEP 2A. Check SDU410 unit power supply for +24-VDC.	Is power supply +24-VDC?
STEP 3.	Check customer interface box wiring.	
	STEP 3A. Check shutdown unit supply and return wires for an open.	Less than 10 ohms?
	STEP 3A-1. Check shutdown unit Modicon™ communication bus supply and return wires for an open.	Less than 10 ohms?

STEPS		SPECIFICATIONS
	STEP 3A-2. Check the battery voltage 1 supply and return wires for an open.	Less than 10 ohms?
	STEP 3B. Check the battery voltage 1 supply and return circuits for supply voltage +24-VDC.	Less than +24-VDC?
	STEP 3C. Check the shutdown unit supply +24-VDC and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 3C-1. Check shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 3C-2. Check the battery voltage 1 supply and return wires for a wire-to-wire short.	Less than 10 ohms?
	STEP 3D. Check the shutdown unit supply +24-VDC wire for a short to ground.	Less than 10 ohms?
	STEP 3D-1. Check shutdown unit Modicon™ communication bus supply wire for a short to ground.	Less than 10 ohms?
	STEP 3D-2. Check the battery voltage 1 supply wire for a short to ground.	Less than 10 ohms?

STEP 1. Check the DCU410 unit.

STEP 1A. Check the power LED for illumination.

Conditions :

- Check the DCU410 unit display.

Action	Specification/Repair	Next Step
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Check the DCU410 unit for illuminated LEDs.	Is the power LED illuminated? YES Repair : Check the DCU410 unit supply and return circuits for intermittent fault. Reference the appropriate troubleshooting tree.	Repair complete
	Is the power LED illuminated? NO Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location for replacement of the DCU410 unit.	2A

STEP 2. Check the DCU410 unit power supply.

STEP 2A. Check the DCU410 unit power supply for +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step

Check the DCU410 unit power supply for +24-VDC.	Is the power supply +24-VDC? YES Repair : Check the batteries. Refer to the OEM service manual.	Repair complete
	Is the power supply +24-VDC? NO Repair : Replace the SDU410 unit. Contact a Cummins® Authorized Repair Location for replacement of the SDU410 unit.	3A

STEP 3. Check the customer interface box wiring.

STEP 3A. Check the shutdown unit supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the supply and return wires at the SDU410 unit.

Action	Specification/Repair	Next Step

Check the shutdown unit supply and return wires for an open. Place one test lead on the shutdown unit supply +24-VDC wire at the SDU410 unit.	Less than 10 ohms? YES	Repair complete
- Place the other test lead on the shutdown unit supply +24-VDC wire at the customer interface box logic unit. - Place one test lead on the shutdown unit supply +24-VDC wire at the SDU410 unit. - Place the other test lead on the shutdown unit supply +24-VDC pin at the X4 connector. - Place one test lead on the shutdown unit return wire at the SDU410 unit. - Place the other test lead on the shutdown unit return wire at the customer interface box logic unit. - Place one test lead on the shutdown unit return wire at the SDU410 unit. - Place the other test lead on the shutdown unit return pin at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	3A-1

STEP 3A-1. Check the shutdown unit Modicon™ communication bus supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply and return wires from the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step
Check the shutdown unit Modicon™ communication bus supply and return wires for an open. • Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. • Place the other test lead on the shutdown unit Modicon™ communication bus supply wire at the DCU410 unit. • Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit. • Place the other test lead on the shutdown unit Modicon™ communication bus return wire at the DCU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	Repair complete
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	3A-2

STEP 3A-2. Check the battery voltage 1 supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the battery voltage 1 supply and return wires at the customer interface box logic unit and the X4 connector.

Action	Specification/Repair	Next Step
Check the battery voltage 1 supply and return wires for an open. • Place one test lead at the battery voltage 1 supply wire at the customer interface box logic unit. • Place the other test lead on the battery voltage 1 supply pin at the X4 connector. • Place one test lead at the battery voltage return wire at the customer interface box logic unit. • Place the other test lead on the battery voltage return pin at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	Repair complete
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	3B

STEP 3B. Check the battery voltage 1 supply and return circuits for supply voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check the battery voltage 1 supply and return circuits for supply voltage +24-VDC. - Place one test lead on the battery voltage 1 supply pin at the X4 connector. - Place the other test lead on the battery voltage 1 return pin at the X4 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Replace the batteries. Refer to the OEM service manual.	3C
	Less than +24-VDC? NO	Repair complete

STEP 3C. Check the shutdown unit supply +24-VDC and return wires for a wire-to-wire short.

Conditions : - Open the customer interface box. - Disconnect the shutdown unit supply +24-VDC and return wires from the SDU410 unit.		
Action	Specification/Repair	Next Step

Check the shutdown unit supply +24-VDC and return wires for a wire-to-wire short. - Place one test lead on the shutdown unit supply +24-VDC wire at the SDU410 unit. - Place the other test lead on all other wires at the SDU410 unit. - Place one test lead on the shutdown unit return wire at the SDU410 unit. - Place the other test lead on all other wires at the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	3C-1

STEP 3C-1. Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short.

Conditions : - Open the customer interface box. - Disconnect the shutdown unit Modicon™ communication bus supply and return wires from the SDU410 unit and DCU410 unit.		
Action	Specification/Repair	Next Step

Check the shutdown unit Modicon™ communication bus supply and return wires for a wire-to-wire short. - Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. - Place the other test lead on all other wires at the SDU410 unit. - Place one test lead on the shutdown unit Modicon™ communication bus return wire at the SDU410 unit. - Place the other test on all other wires at the SDU410 unit.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	3C-2

STEP 3C-2. Check the battery voltage 1 supply and return wires for a wire-to-wire short.

Conditions : - Open the customer interface box. - Disconnect the battery voltage 1 supply and return wires from the SDU410 unit and DCU410 unit.		
Action	Specification/Repair	Next Step

Check the battery voltage 1 supply and return wires for a wire-to-wire short. • Place one test lead on the battery voltage 1 supply wire at the SDU410 unit. • Place the other test lead on all other wires at the SDU410 unit. • Place one test lead on the battery voltage 1 return wire at the SDU410 unit. • Place the other test lead on all other wires at the SDU410 unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	3D

STEP 3D. Check the shutdown unit supply +24-VDC wire for a short to ground.

Conditions : • Open the customer interface box. • Disconnect the shutdown unit supply +24-VDC wire from the SDU410 unit.		
Action	Specification/Repair	Next Step

Check the shutdown unit supply +24-VDC wire for a short to ground. - Place one test lead on the shutdown unit supply +24-VDC at the SDU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	3D-1

STEP 3D-1. Check the shutdown unit Modicon™ communication bus supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the shutdown unit Modicon™ communication bus supply wire from the SDU410 unit and DCU410 unit.

Action	Specification/Repair	Next Step
Check the shutdown unit Modicon™ communication bus supply wire for a short to ground. - Place one test lead on the shutdown unit Modicon™ communication bus supply wire at the SDU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	3D-2

STEP 3D-2. Check the battery voltage 1 supply wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the battery voltage 1 supply wire from the customer interface box logic unit and X4 connector.

Action	Specification/Repair	Next Step
Check the battery voltage 1 supply wire for a short to ground. • Place one test lead on the battery voltage 1 supply wire at the customer interface box logic unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete
	Less than 10 ohms? NO	Contact a Cummins® Authorized Repair Location